

# BEYOND THE FORD-CIRCLE LAW: PARABOLIC RENORMALIZATION AND THE SHAPE OF MANDELBROT SATELLITE BULBS

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ABSTRACT. To first order, the satellite component of the Mandelbrot set attached to the main cardioid at internal angle  $p/q$  is a round disc of diameter proportional to  $q^{-2}$ ; this is the Ford-circle law. We study the first correction. Expanding the multiplier of the bifurcating  $q$ -cycle in the bulb-scale parameter  $u = q^2\varepsilon$ , we prove that the first coefficient depending on  $p/q$  is  $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$ , where  $\iota_{p/q}$  is the holomorphic index of the parabolic fixed point of  $f^q$ . We also prove that  $\kappa(p/q) + \frac{1}{2}$  is the index of the rotation quotient of  $f^q$ , and we give an exact  $O(q^2)$  recursion for it. We then present numerical evidence, at up to 24 digits, that the correction is governed by parabolic renormalization. The limit  $\kappa_0 = \lim \kappa(1/q)$  equals  $\iota(\mathcal{P}_0) - \frac{1}{2}$ , where  $\mathcal{P}_0$  is the parabolic renormalization of  $z + z^2$ . Along bounded  $p$ , the limits of  $\kappa$  and of the bulb diameter are those of  $\mathcal{P}_0$  times the Inou–Shishikura phase. Numerically, the entire  $O(1/q)$  correction is kinematic: it is produced by the exact phase, which is a Möbius function of  $u$ . Along continued fractions with large partial quotients the constants form a hierarchy that converges geometrically, with ratio 0.042, to a constant  $\kappa_*$  attached to parabolic renormalization composed with complex conjugation; its fixed point differs from the one computed by Lanford and Yampolsky. At every rational  $p/q$ , the bulbs accumulating at the root of  $W_{p/q}$  from above and from below are governed by the upper and lower parabolic renormalizations of  $e^{2\pi ip/q}z + z^2$ , so  $\kappa$  has distinct one-sided limits at every rational; the period-two disc is governed by the same germ at its root. For  $z^3 + c$  the picture holds with the cubic cusp germ. We compute three new constants to 30 digits.

## 1. INTRODUCTION

**1.1. The Ford-circle law and its first correction.** Let  $\lambda \in \mathbb{D}$  parametrize the main cardioid of the Mandelbrot set  $M$  by the multiplier of the attracting fixed point,  $c(\lambda) = \lambda/2 - \lambda^2/4$ . Equivalently we work with

$$f_\lambda(z) = \lambda z + z^2,$$

which is affinely conjugate to  $z^2 + c(\lambda)$ . At each root of unity  $\lambda_0 = e^{2\pi ip/q}$ ,  $\gcd(p, q) = 1$ , a satellite hyperbolic component  $W_{p/q}$  of period  $q$  is attached to the cardioid [2]. Its size is of order  $q^{-2}$  [7]. More precisely, to first order in  $1/q$  it is a round disc of diameter  $2|c'(\lambda_0)|q^{-2} = 2\sin(\pi p/q)q^{-2}$  [5]. Up to the factor  $\sin(\pi p/q)$  these are the radii of the Ford circles [6], and the satellites are arranged in the Farey order [11]; this is the leading-order content of the familiar analogy between  $M$  and the Farey tessellation.

This paper studies the *first correction* to that law, which to our knowledge has not been investigated before. Put  $\lambda = \lambda_0 e^\varepsilon$  and let  $\rho_{p/q}(\varepsilon)$  be the multiplier of the  $q$ -cycle of  $f_\lambda$  that

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*Date:* September 2026.

*2020 Mathematics Subject Classification.* Primary 37F46; Secondary 37F25, 37F10, 11A55.

*Key words and phrases.* Mandelbrot set, satellite components, Ford circles, holomorphic index, résidu itératif, parabolic implosion, horn map, near-parabolic renormalization.

bifurcates from the fixed point 0. In the bulb-scale variable  $u = q^2\varepsilon$  write

$$(1) \quad R_{p/q}(u) := \rho_{p/q}(u/q^2) = \sum_{k \geq 0} r_k(p/q) u^k.$$

Up to the conformal change of variable  $u \mapsto c(\lambda_0 e^{u/q^2})$ , the component  $W_{p/q}$  is the set  $\{|R_{p/q}(u)| < 1\}$ . We show that  $r_0 = 1$  and  $r_1 = -1$  for every  $p/q$ , so that to first order  $W_{p/q}$  is the disc  $|u - 1| < 1$ : this is the Ford law. The first coefficient that depends on  $p/q$  is

$$\kappa(p/q) := r_2(p/q),$$

and it governs the leading deviation of  $W_{p/q}$  from a round disc. When  $\lambda = e^{2\pi i \alpha}$  we also write  $f_{e^{2\pi i \alpha}}$  for  $f_\lambda$ ; in particular  $f_1(z) = z + z^2$ .

**1.2. Exact results.** Let  $f = f_{\lambda_0}$ , so that  $f^q(z) = z + a z^{q+1} + O(z^{q+2})$  with  $a \neq 0$ , and let  $\iota_{p/q} = \text{Res}_{z=0} dz/(z - f^q(z))$  be the holomorphic index of 0 as a fixed point of  $f^q$ .

- **Index formula** (Theorem 2.1).  $\rho_{p/q}(\varepsilon) = 1 - q^2\varepsilon + q^3(\iota_{p/q} - \frac{1}{2})\varepsilon^2 + O(\varepsilon^3)$ , that is,  $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$ . In particular  $\kappa(p/q) \in \mathbb{Q}(\zeta_q)$ , and the values  $\kappa(p'/q)$ ,  $\gcd(p', q) = 1$ , form a single Galois orbit. The index formula holds for the satellites of every hyperbolic component of  $z^d + c$ .
- **Normal form** (Theorem 3.1, Corollary 3.2). If  $\mathcal{N}(z) = \lambda_0 z(1 + b_1 z^q + b_2 z^{2q})$  is the resonant normal form of  $f$ , then  $q \iota_{p/q} = \frac{q^2-1}{2} + b_2/b_1^2$  and  $\kappa(p/q) = \iota(Q_{p/q}) - \frac{1}{2}$ , where  $Q_{p/q}(W) = \mathcal{N}^q(W^{1/q})^q$  is the  $q$ -fold rotation quotient of the normal form of  $f^q$ . The coefficients  $b_1, b_2$  come from an explicit recursion with  $O(q^2)$  operations.
- **Horn-map indices** (Lemma 4.3). If  $E(Z) = Z + a_0 + \sum_{n \geq 1} a_n e^{2\pi i n Z}$  is an upper horn map and  $\mathcal{P}(W) = W \exp(2\pi i \sum_n a_n W^n)$  the corresponding parabolic germ, then  $\iota(\mathcal{P}) - \frac{1}{2} = a_2/(2\pi i a_1^2)$ .
- **Kinematics** (Proposition 5.1). If  $\lambda = e^{2\pi i \alpha} = \lambda_0 e^{u/q^2}$ , then exactly  $e^{-2\pi i / \alpha} = e^{-2\pi i q/p} e^{\tilde{u}/p^2}$ , where  $\tilde{u} = u/(1 + \frac{u}{2\pi i p q})$  is a Möbius function of  $u$ .

**1.3. Numerical results and conjectures.** The rest of the paper supports one picture: up to  $O(q^{-2})$ , the bulb  $W_{p/q}$  is described by a near-parabolic renormalization of  $f_{\lambda_0}$  in the sense of Inou and Shishikura [8], composed with the exact phase of Proposition 5.1 (Conjecture 5.2). The main consequences, each tested against the bulbs, are listed below. Table 1 collects the constants.

- (1) *The constant  $\kappa_0$*  (§4).  $\kappa(1/q) \rightarrow \kappa_0 = \iota(\mathcal{P}_0) - \frac{1}{2}$ , where  $\mathcal{P}_0$  is the parabolic renormalization of  $z + z^2$ . The extrapolation of values certified up to  $q = 32768$  agrees with the horn map to at least 24 digits ( $3 \cdot 10^{-28}$  on the best fit).
- (2) *Bounded  $p$  and the bulb size* (§5). Along  $q = pN + r$  the limits of  $\kappa$  and of the bulb diameter are those of the family  $e^{-2\pi i r/p} e^{u/p^2} \mathcal{P}_0$ . In all classes with  $p \leq 5$  they agree within the extrapolation uncertainties, which are about  $10^{-7}$  for  $\kappa$  and  $10^{-6}$  for the diameter.
- (3) *Kinematics* (§5.4). Numerically,  $R_{p/q}(u) = \rho_{p,r}(\tilde{u}(u)) + O(q^{-2})$  coefficientwise. This accounts for the universal term  $-i/(2\pi p q)$  in  $\kappa$  and for the whole  $O(1/q)$  part of  $R_{p/q}$ .
- (4) *The hierarchy* (§6). Iterating along continued fractions with large partial quotients gives constants  $\kappa_k$  that contract by the factor 0.042 per level towards  $\kappa_*$ . The levels  $k \leq 3$  are confirmed on bulbs, the third to about  $10^{-5}$ . The fixed point differs from that of Lanford and Yampolsky [13], whose computations our code reproduces.

- (5) *Limits at the roots of satellites* (§7). As the internal angle tends to a rational  $p/q$  with the last partial quotient tending to infinity,  $\kappa$  tends to the index of the upper (from above) or lower (from below) parabolic renormalization of  $e^{2\pi ip/q}z + z^2$ , and the whole bounded-tail theory carries over. For  $q \geq 3$  the two one-sided limits are not conjugate; eleven cases at  $1/3$ ,  $1/4$ ,  $2/5$  and  $3/7$  agree to  $6 \cdot 10^{-11}$ . The satellites of the period-two disc near its root follow the same germ ( $q = 2$ ), with its own constant  $\kappa_{1/2}$ . We conjecture that in general only the root germ matters.
- (6) *Degree three* (§8). For  $z^3 + c$  the same predictions hold numerically with the cusp germ  $v + v^2 + \frac{1}{3}v^3$ ; the bulbs now have two antipodes (Proposition 8.1).

We state as conjectures only what the evidence supports quantitatively: Conjectures 4.5, 5.2, 6.3 and 7.3. Values of  $\kappa$  are computed either in certified ball arithmetic or at two working precisions. Bulb data away from the root (bulb sizes, the disc, degree three) are computed in double precision and checked by the identity  $r_1 = -1$ . Limits are extrapolations, and we give their uncertainty as the spread of stable fits (Appendix A).

**1.4. Related work.** The  $q^{-2}$  law is due to Guckenheimer and McGehee [7], and the leading constant  $2 \sin(\pi p/q)$  was derived by Fowler and McGuinness [5]. Parabolic implosion and horn maps go back to Douady, Lavaurs and Shishikura [14, 19]. Near-parabolic renormalization and its invariant class are due to Inou and Shishikura [8], and Lanford and Yampolsky [13] studied the fixed point of parabolic renormalization numerically. Near-parabolic renormalization for unicritical maps of higher degree was developed by Chéritat [10] and, for local degree three, by Yang [20]. Buff and Chéritat [9] used parabolic implosion to control the size of quadratic Siegel discs in terms of the Brjuno function, which is the closest precedent we know for quantitative asymptotics of this kind. The holomorphic index and the résidu itératif are classical [15, 3], and Buff and Epstein [1] proved an inequality of Pommerenke–Levin–Yoccoz type for parabolic cycles, which by Theorem 2.1 constrains  $\kappa$ ; we do not pursue this here. The Ford-circle analogy is usually stated only to leading order [11]. We are not aware of previous work on the corrections, and we have not found Theorem 2.1 in the literature, although it follows directly from the holomorphic fixed-point formula.

**1.5. Organization.** Section 2 proves the index formula, and Section 3 gives the normal form and the rotation quotient. Section 4 treats the horn map of  $z + z^2$  and  $\kappa_0$ . Section 5 formulates the renormalization principle and tests it at bounded  $p$  and at first order in  $1/q$ . Sections 6 and 7 treat the hierarchy and the dependence on the component, and Section 8 treats degree three. Section 9 reports integer-relation searches, and Section 10 lists open problems. The appendix describes the numerical methods.

## 2. THE INDEX FORMULA

Fix  $p/q$  in lowest terms,  $\lambda_0 = e^{2\pi ip/q}$ , and write  $f = f_{\lambda_0}$ . Since every cycle of petals attracts a critical point [15, §10] and  $f$  has only one, the parabolic fixed point 0 has exactly one cycle of  $q$  attracting petals, so

$$(2) \quad f^q(z) = z + a z^{q+1} + O(z^{q+2}), \quad a = a(p/q) \neq 0.$$

The holomorphic index [15, §12] of 0 as a fixed point of  $f^q$  is

$$\iota_{p/q} := \frac{1}{2\pi i} \oint \frac{dz}{z - f^q(z)} = \operatorname{Res}_{z=0} \frac{dz}{z - f^q(z)},$$

and its résidu itératif [3, 1] is  $\operatorname{résit}(f^q) = \frac{q+1}{2} - \iota_{p/q}$ . In general, for a fixed point of multiplier one and multiplicity  $m+1$  we set  $\operatorname{résit} = \frac{m+1}{2} - \iota$ .

| constant         | germ                                               | value                                         | bulbs                              |      |
|------------------|----------------------------------------------------|-----------------------------------------------|------------------------------------|------|
| $\kappa_0$       | $z + z^2$                                          | $0.02382588740220057 - 0.05230465911400367 i$ | $3 \cdot 10^{-28}$                 | §4   |
| $\kappa_0^{(3)}$ | $v + v^2 + \frac{1}{3}v^3$                         | $0.1435992780952916 - 0.03031164789850254 i$  | $1.3 \cdot 10^{-6}$ ( $q = 1024$ ) | §8   |
| $\kappa_{1/2}$   | $-z + z^2$                                         | $0.01840526161796478 - 0.04290372388023048 i$ | $3 \cdot 10^{-10}$                 | §7   |
| $\kappa_*$       | $\mathcal{G}_* = \mathcal{P}(\bar{\mathcal{G}}_*)$ | $0.011929183413 - 0.037455493752 i$           | levels $\leq 3$                    | §6   |
| $G_0$            | $e^u \mathcal{P}_0$                                | 1.02641539                                    | $4 \cdot 10^{-8}$                  | §5.3 |

TABLE 1. The constants of this paper. The column “bulbs” gives the difference between the horn-map value and the best extrapolated bulb limit; for  $\kappa_0^{(3)}$  it gives the error at  $q = 1024$ , which decays like  $q^{-2}$ . The extrapolation uncertainties are larger: about  $3 \cdot 10^{-25}$  for  $\kappa_0$ ,  $10^{-7}$  for  $\kappa_{1/2}$  and  $10^{-6}$  for  $G_0$ .

**Theorem 2.1** (Index formula). *The multiplier  $\rho_{p/q}(\varepsilon)$  of the  $q$ -cycle of  $f_{\lambda_0 e^\varepsilon}$  that bifurcates from 0 is holomorphic near  $\varepsilon = 0$ , and*

$$\rho_{p/q}(\varepsilon) = 1 - q^2 \varepsilon + q^3 \left( \iota_{p/q} - \frac{1}{2} \right) \varepsilon^2 + O(\varepsilon^3).$$

Equivalently, in (1),  $r_0 = 1$ ,  $r_1 = -1$  and  $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$ .

*Proof.* Choose a disc  $D$  around 0 on whose boundary  $z \neq f^q(z)$ , and so small that 0 is the only fixed point of  $f^\ell$  in  $\bar{D}$  for every  $\ell \leq q$ ; by continuity the second property persists for  $f_\lambda^\ell$ ,  $\ell < q$ , when  $\lambda$  is close to  $\lambda_0$ . By (2) and Rouché’s theorem, for  $|\varepsilon|$  small the map  $f_\lambda^q$ ,  $\lambda = \lambda_0 e^\varepsilon$ , has exactly  $q + 1$  fixed points in  $D$ , counted with multiplicity. One of them is 0, with multiplier  $\lambda^q = e^{q\varepsilon} \neq 1$ , so it is simple. The remaining  $q$  points therefore have exact period  $q$ . They are permuted by  $f_\lambda$ , since  $f_\lambda$  maps fixed points of  $f_\lambda^q$  near 0 to fixed points of  $f_\lambda^q$  near 0, and they form a single cycle because  $f_\lambda$  is close to the rotation by  $\lambda_0$  near 0; call its multiplier  $\rho(\varepsilon) = \prod_j (\lambda + 2z_j)$ . The power sums  $\sum_j z_j^k = \frac{1}{2\pi i} \oint_{\partial D} z^k \frac{1 - (f_\lambda^q)'(z)}{z - f_\lambda^q(z)} dz$ ,  $k \geq 1$ , are holomorphic in  $\lambda$ , and hence so is  $\rho$ , a symmetric polynomial in the  $z_j$ . The function  $\rho$  is not identically 1, since the cycle is attracting on the satellite, so for  $\varepsilon \neq 0$  small all  $q + 1$  fixed points are simple. Summing their indices  $1/(1 - m_j)$ , where  $m_j$  are the multipliers, gives

$$(3) \quad \frac{1}{1 - e^{q\varepsilon}} + \frac{q}{1 - \rho(\varepsilon)} = \frac{1}{2\pi i} \oint_{\partial D} \frac{dz}{z - f_\lambda^q(z)} \xrightarrow{\varepsilon \rightarrow 0} \iota_{p/q},$$

because the integrand converges uniformly on  $\partial D$ . The first term equals  $-1/(q\varepsilon) + \frac{1}{2} + O(\varepsilon)$  and is unbounded. Hence  $\rho(0) = 1$ . Writing  $1 - \rho = A\varepsilon^n(1 + O(\varepsilon))$  with  $A \neq 0$ , boundedness of the sum forces  $n = 1$  and  $A = q^2$ . With  $\rho = 1 - q^2\varepsilon + C\varepsilon^2 + O(\varepsilon^3)$  we get  $q/(1 - \rho) = 1/(q\varepsilon) + C/q^3 + O(\varepsilon)$ , and (3) yields  $\frac{1}{2} + C/q^3 = \iota_{p/q}$ .  $\square$

*Remark 2.2* (Other components). The proof uses only that the parabolic point has multiplicity  $q + 1$ . It therefore applies verbatim to the satellites of any hyperbolic component of  $z^d + c$ , whose parabolic cycles have a single cycle of petals, with  $\lambda$  the multiplier of the attracting cycle of that component (so that the first return map has multiplier exactly  $\lambda$  at the cycle point) and  $f^q$  replaced by the  $q$ -th iterate of the first return map. The arithmetic statements of Remark 2.3 use that  $f_{\lambda_0}$  has coefficients in  $\mathbb{Z}[\lambda_0]$  and do not extend verbatim. We checked numerically, for the period-two disc and for  $z^3 + c$ , that  $r_2 = (\iota - \frac{1}{2})/q$  holds to  $10^{-7}$  at  $p/q = 2/7$  and  $3/11$ .

*Remark 2.3* (Exact values). Since  $f^q$  has coefficients in  $\mathbb{Z}[\lambda_0]$ , the index is a rational function of  $\lambda_0$  with rational coefficients. Hence  $\iota_{p/q} \in \mathbb{Q}(\zeta_q)$  and  $\iota_{p'/q} = \sigma(\iota_{p/q})$  for the automorphism  $\sigma$  :

$\zeta_q \mapsto \zeta_q^{p/\bar{p}}$ , where  $p\bar{p} \equiv 1 \pmod{q}$ . For example  $\iota_{1/2} = \frac{1}{8}$ ,  $\iota_{1/3} = \frac{100-8\sqrt{3}i}{441}$  and  $\iota_{1/4} = \frac{1447-365i}{4624}$ , so that  $\kappa(1/2) = -\frac{3}{16}$  and  $\kappa(1/3) = \frac{-241-16\sqrt{3}i}{2646}$ .

**Definition 2.4.** Let  $c_{\text{root}} = c(\lambda_0)$ , and let  $c_{\text{ant}} \in \partial W_{p/q}$  be the boundary point at which the  $q$ -cycle has multiplier  $-1$ . It is unique because in degree two the multiplier map extends to a homeomorphism  $\overline{W}_{p/q} \rightarrow \overline{\mathbb{D}}$  [2]. Set

$$\delta(p/q) = |c_{\text{ant}} - c_{\text{root}}|, \quad G(p/q) = \frac{\delta(p/q)}{2|c'(\lambda_0)|q^{-2}}.$$

Since  $c'' \equiv -\frac{1}{2}$  and  $|c'(\lambda_0)| = \frac{1}{2}|1 - \lambda_0| = \sin(\pi p/q)$ , we have  $c(\lambda_0 e^{u/q^2}) - c(\lambda_0) = c'(\lambda_0)\lambda_0 \frac{u}{q^2} (1 + O(\frac{|u|}{q^2 \sin(\pi p/q)}))$ , and therefore  $G(p/q) = \frac{1}{2}|u_a|(1 + O(\frac{1}{q^2 \sin(\pi p/q)}))$ , where  $R_{p/q}(u_a) = -1$ . For bounded  $p$  the error is  $O(1/(pq))$ . To first order  $u_a = 2$  and  $G = 1$ ; the deviation of  $G$  from 1 depends on all the  $r_k$ , of which  $\kappa$  is the first.

### 3. NORMAL FORMS AND THE ROTATION QUOTIENT

**Theorem 3.1** (Normal form). *There is a unique polynomial  $H(z) = z + \sum_{k=2}^{2q} H_k z^k$  with  $H_{q+1} = 0$  such that  $f \circ H = H \circ \mathcal{N}$  up to order  $2q+1$ , where*

$$\mathcal{N}(z) = \lambda_0 z (1 + b_1 z^q + b_2 z^{2q}).$$

Put  $\hat{b}_i = \lambda_0 b_i$  and let  $(H^2)_k$  denote the coefficient of  $z^k$  in  $H(z)^2$ . The coefficients are determined by the recursion

$$(4) \quad H_k(\lambda_0 - \lambda_0^k) = [k > q+1] (k-q)\lambda_0^{k-q-1} \hat{b}_1 H_{k-q} - (H^2)_k \quad (2 \leq k \leq 2q, k \neq q+1),$$

together with  $\hat{b}_1 = (H^2)_{q+1}$  and  $\hat{b}_2 = (H^2)_{2q+1}$ . Moreover

$$a(p/q) = q b_1, \quad q \iota_{p/q} = \frac{q^2 - 1}{2} + \frac{b_2}{b_1^2}, \quad \kappa(p/q) = \frac{1}{2} - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2}.$$

*Proof.* Comparing coefficients of  $z^k$  in  $\lambda_0 H + H^2 = \sum_m H_m \mathcal{N}^m$  gives (4). Here we use that  $[\mathcal{N}^m]_k$  is nonzero only for  $k \in \{m, m+q, m+2q, \dots\}$ , with values  $\lambda_0^m$  and  $m\lambda_0^{m-1}\hat{b}_1$  in the first two cases. The factor  $\lambda_0 - \lambda_0^k$  vanishes exactly when  $k \equiv 1 \pmod{q}$ . At  $k = q+1$  and  $k = 2q+1$  the equation fixes  $\hat{b}_1$  and  $\hat{b}_2$  and leaves  $H_{q+1}$  free; we set it to zero. For the second part, write  $\mathcal{N}^j(z) = \lambda_0^j z (1 + A_j z^q + B_j z^{2q} + \dots)$ . Using  $\lambda_0^q = 1$  and  $(\mathcal{N}^j)^q = z^q (1 + qA_j z^q + \dots)$ ,

$$A_{j+1} = A_j + b_1, \quad B_{j+1} = B_j + (q+1)b_1 A_j + b_2,$$

so  $A_q = qb_1$  and  $B_q = \frac{(q+1)q(q-1)}{2} b_1^2 + qb_2$ . The coefficient  $a$  and the index are invariant under formal conjugacy tangent to the identity, and for  $z + Az^{q+1} + Bz^{2q+1}$  the index is  $B/A^2$ . This gives the formulas for  $a$  and  $\iota$ , and the formula for  $\kappa$  follows from Theorem 2.1.  $\square$

**Corollary 3.2** ( $\kappa$  is an index). *Since  $\mathcal{N}$  commutes with  $z \mapsto \lambda_0 z$ , the rotation quotient  $Q_{p/q}(W) := \mathcal{N}^q(W^{1/q})^q = W + q^2 b_1 W^2 + \dots$  is a well-defined germ, and*

$$\kappa(p/q) = \iota(Q_{p/q}) - \frac{1}{2} = \frac{1}{2} - \text{résit}(Q_{p/q}).$$

*Proof.* From  $\mathcal{N}^q(z) = z(1 + A_q z^q + B_q z^{2q} + \dots)$  we get  $Q_{p/q}(W) = W(1 + A_q W + B_q W^2)^q + \dots$ . Hence  $Q_{p/q} = W + AW^2 + BW^3 + \dots$  with  $A = qA_q = q^2 b_1$  and  $B = qB_q + \binom{q}{2} A_q^2$ . Then  $\iota(Q_{p/q}) = B/A^2 = 1 - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2}$ , which is  $\kappa + \frac{1}{2}$  by Theorem 3.1.  $\square$

*Remark 3.3* (Relation with the résidu itératif). Only the 3-jet of  $Q_{p/q}$  is determined by  $f$ ; the higher coefficients depend on the truncation of the normal form, and all that enters  $\iota(Q_{p/q})$  is the 3-jet. In terms of the résidu itératif, Theorem 2.1 reads  $\kappa(p/q) = \frac{1}{2} - \text{résit}(f^q)/q$ . Corollary 3.2 is equivalent to it: with  $Q_1(W) = \mathcal{N}(W^{1/q})^q$  one has  $Q_{p/q} = Q_1^q$ , a direct computation from Theorem 3.1 gives  $\text{résit}(Q_1) = \frac{q+1}{2q} - \frac{b_2}{qb_1^2} = \text{résit}(f^q)$ , and  $\text{résit}(g^q) = \text{résit}(g)/q$  for any germ  $g(W) = W + AW^2 + BW^3 + \dots$  with  $A \neq 0$ . The content of Theorem 3.1 is the recursion.

*Remark 3.4* (Computation). The recursion (4) costs  $O(q^2)$  arithmetic operations. It is numerically stable: at  $q = 128$  and  $q = 301$  the values of  $\kappa(1/q)$  computed at 60 and at over 300 working digits agree to more than twenty digits, and for generic  $p$  with  $q$  up to  $6.4 \cdot 10^4$  two runs at 128 and 192 bits agree to  $4 \cdot 10^{-35}$ . By contrast, composing  $f$  symbolically  $q$  times costs  $O(q^3)$  and suffers severe cancellation: the coefficients of  $f^q$  are huge (for instance  $|a(1/q)|$  grows like  $(q/2\pi)^{q+o(q)}$ , and  $|a(1/256)| \approx 10^{414}$ ), and the index computed that way needed about  $1.5q$  working digits. With a blocked online convolution the recursion runs in ball arithmetic [12], which certifies  $\kappa(1/q)$  for  $q \leq 32768$  (Appendix A).

#### 4. THE HORN MAP OF $z + z^2$ AND THE CONSTANT $\kappa_0$

**4.1. The limit along  $1/q$ .** Table 2 and Figure 1 show  $\kappa(1/q)$  for  $q$  up to 32768.

**Numerical result 4.1.** Least-squares fits  $\kappa(1/q) = \sum_{j \leq J} \gamma_j q^{-j}$  over  $q \geq q_{\min}$  are stable under changes of  $(q_{\min}, J)$ . The best overdetermined fit,  $(q_{\min}, J) = (256, 10)$  on thirteen values with residual  $3.6 \cdot 10^{-29}$ , gives

$$\gamma_0 = 0.023825887402200569000729189339 \dots - 0.052304659114003666316522486209 \dots i,$$

the first-order coefficient  $\gamma_1 = -i/(2\pi)$  to within  $2 \cdot 10^{-23}$ , and  $\gamma_2 = 0.402776138694 + 2.902726170410i$ . The fits with  $(q_{\min}, J) = (128, 10)$ ,  $(256, 9)$  and  $(512, 9)$  give  $\gamma_0$  within  $3 \cdot 10^{-25}$  of this value. The ratios  $|\gamma_{j+1}/\gamma_j|$  (2.8, 18.4, 6.3, 6.9,  $\dots$ , 8.2) drift slowly upward, so the expansion may be asymptotic rather than convergent; we use the fits only as extrapolations.

**4.2. Fatou coordinates and the horn map.** We recall the objects of parabolic implosion [14, 19, 8, 13] in the form in which we compute them. In the coordinate  $w = -1/z$  the map  $f_1(z) = z + z^2$  becomes

$$F(w) = \frac{w^2}{w-1} = w + 1 + \frac{1}{w-1}.$$

**Lemma 4.2** (Formal Fatou coordinate). *There is a unique formal series  $\Phi(w) = w - \log w + \sum_{k \geq 1} c_k w^{-k}$  without constant term such that  $\Phi \circ F = \Phi + 1$ . Its coefficients are rational,*

$$c_1, c_2, \dots = \frac{1}{2}, \frac{1}{3}, \frac{13}{36}, \frac{113}{240}, \frac{1187}{1800}, \frac{877}{945}, \dots,$$

and with  $t = 1/w$  they satisfy the triangular system

$$\sum_{k \geq 1} c_k [(t - t^2)^k - t^k] = - \sum_{m \geq 2} \left(1 - \frac{1}{m}\right) t^m.$$

*Proof.* With  $t = 1/w$  we have  $1/F = t - t^2$ ,  $F - w - 1 = t/(1 - t)$  and  $\log F - \log w = -\log(1 - t)$ . Substituting gives the displayed identity. The coefficient of  $t^m$  contains  $c_{m-1}$  with the factor  $-(m-1)$  and otherwise only  $c_k$  with  $k < m-1$ . The coefficient of  $\log w$  is fixed by the term of order  $t$ ; it equals the résidu itératif 1 of  $f_1$ .  $\square$

| $q$                         | $\kappa(1/q)$                                 | $ \kappa(1/q) + \frac{i}{2\pi q} - \kappa_0 $ |
|-----------------------------|-----------------------------------------------|-----------------------------------------------|
| 64                          | 0.02385454027565757 - 0.05408067667475267 $i$ | $7.1 \cdot 10^{-4}$                           |
| 256                         | 0.02383094323212187 - 0.05288194473715776 $i$ | $4.5 \cdot 10^{-5}$                           |
| 1024                        | 0.02382625451962614 - 0.05245731336604759 $i$ | $2.8 \cdot 10^{-6}$                           |
| 4096                        | 0.02382591114406885 - 0.05234334224894280 $i$ | $1.7 \cdot 10^{-7}$                           |
| 8192                        | 0.02382589337085374 - 0.05232404394907701 $i$ | $4.4 \cdot 10^{-8}$                           |
| 16384                       | 0.02382588889851146 - 0.05231436234675194 $i$ | $1.1 \cdot 10^{-8}$                           |
| 32768                       | 0.02382588777679670 - 0.05230951343396723 $i$ | $2.7 \cdot 10^{-9}$                           |
| fit, $q \rightarrow \infty$ | 0.02382588740220057 - 0.05230465911400367 $i$ |                                               |
| horn map, Num. res. 4.4     | 0.02382588740220057 - 0.05230465911400367 $i$ |                                               |

TABLE 2.  $\kappa(1/q)$  from the normal-form recursion (first sixteen digits shown). The corrected error decays like  $q^{-2}$ . Values for  $q \geq 1024$  are certified in ball arithmetic (radius  $\leq 5.4 \cdot 10^{-44}$ ). The fit uses the thirteen values  $256 \leq q \leq 32768$  with eleven terms in  $1/q$  (residual  $3.6 \cdot 10^{-29}$ ). The last row is the horn-map value of Numerical result 4.4.

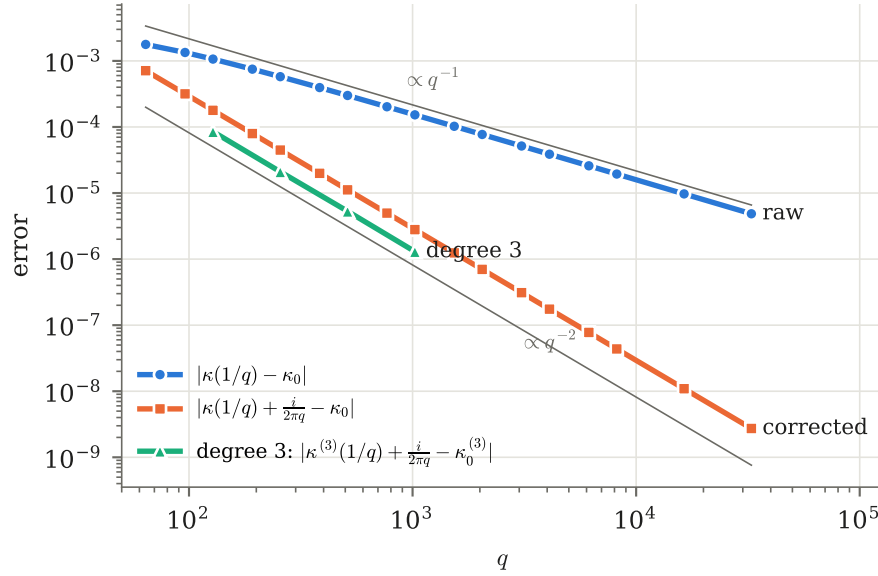


FIGURE 1. Convergence of  $\kappa(1/q)$  to  $\kappa_0$  in degree two and of  $\kappa^{(3)}(1/q)$  to  $\kappa_0^{(3)}$  in degree three (Section 8). The raw error decays like  $q^{-1}$ ; after the universal correction  $-i/(2\pi q)$  it decays like  $q^{-2}$ .

Let  $\Phi_{\text{att}}(w) = \lim_n (\Phi(F^n(w)) - n)$  on the attracting petal, with the principal logarithm, and let  $\Psi_{\text{rep}}(Z) = \lim_n F^n(\Phi^{-1}(Z - n))$  be the repelling Fatou parametrization, with the branch  $\arg w \in (0, 2\pi)$ . The upper horn map  $E = \Phi_{\text{att}} \circ \Psi_{\text{rep}}$  commutes with  $Z \mapsto Z + 1$  and, for  $\text{Im } Z$  large, has an expansion

$$E(Z) = Z + a_0 + \sum_{n \geq 1} a_n e^{2\pi i n Z}.$$

For  $\text{Im } Z > 0$  the two branches of the logarithm agree, so both Fatou coordinates have the same asymptotic expansion there and  $a_0 = 0$  with these normalizations. Translating both  $\Phi_{\text{att}}$  and  $\Psi_{\text{rep}}$  by  $\tau$  conjugates  $E$  by  $Z \mapsto Z + \tau$ , which sends  $a_n \mapsto a_n e^{2\pi i n \tau}$ ; translating only one of them changes  $a_0$ . In  $W = e^{2\pi i Z}$  the horn map is the germ  $W \mapsto e^{2\pi i a_0} \mathcal{P}_0(W)$ , where

$$\mathcal{P}_0(W) := W \exp\left(2\pi i \sum_{n \geq 1} a_n W^n\right)$$

is the parabolic renormalization of  $f_1$ , normalized to multiplier one [8, 13]. More generally, let  $g(v) = v + v^2 + O(v^3)$  be holomorphic near 0. The same construction in  $w = -1/v$ , with  $\Phi(w) = w - \text{résit}(g) \log w + \sum c_k w^{-k}$  and the same branches, gives an upper horn map and a germ  $W \exp(2\pi i \sum_n a_n W^n)$ . Conjugating it by  $W \mapsto 2\pi i a_1 W$  we normalize it to the form  $v + v^2 + O(v^3)$ , and we call the result  $\mathcal{P}(g)$ . Thus  $\mathcal{P}_0$  and  $\mathcal{P}(f_1)$  differ only by this linear conjugacy.

**Lemma 4.3.** *If  $a_1 \neq 0$ , then  $\iota(\mathcal{P}_0) = \frac{1}{2} + \frac{a_2}{2\pi i a_1^2}$ . This quantity does not depend on the normalizations of the Fatou coordinates, and the same formula holds for  $\mathcal{P}(g)$ .*

*Proof.*  $\mathcal{P}_0(W) = W + AW^2 + BW^3 + \dots$  with  $A = 2\pi i a_1$  and  $B = 2\pi i a_2 + \frac{1}{2}(2\pi i a_1)^2$ , and  $\iota = B/A^2$ . Under  $a_n \mapsto a_n e^{2\pi i n \tau}$  the ratio  $a_2/a_1^2$  is unchanged,  $a_0$  does not enter, and  $\iota$  is invariant under conjugacy.  $\square$

**Numerical result 4.4.** Computing  $E$  on the line  $\text{Im } Z = \frac{1}{4}$  at 60 and at 80 digits (Appendix A) gives  $|a_0| < 10^{-58}$ , as it must with these branches, and

$$\begin{aligned} \text{Re } a_1 &= -0.05895043056440687000807026251551, \\ \text{Im } a_1 &= 0.010009865898273724511535250164253, \\ \text{Re } a_2 &= 0.0012858196663089284889517464706577, \\ \text{Im } a_2 &= 0.00011738770718282661923962731179377, \end{aligned}$$

with  $|a_n/a_{n+1}|$  increasing to about 55.6. Hence

$$\begin{aligned} \text{Re}(\iota(\mathcal{P}_0) - \tfrac{1}{2}) &= 0.023825887402200569000729189645402, \\ \text{Im}(\iota(\mathcal{P}_0) - \tfrac{1}{2}) &= -0.052304659114003666316522486207571. \end{aligned}$$

The two precision settings agree in all digits shown. This value coincides with  $\gamma_0$  of Numerical result 4.1 to  $3.1 \cdot 10^{-28}$ , and the other stable fits differ from it by at most  $3 \cdot 10^{-25}$ . We therefore claim agreement to at least 24 digits.

**Conjecture 4.5.** The limit  $\kappa_0 = \lim_{q \rightarrow \infty} \kappa(1/q)$  exists and equals  $\iota(\mathcal{P}_0) - \frac{1}{2} = \frac{a_2}{2\pi i a_1^2}$ . Moreover  $\kappa(1/q) = \kappa_0 - \frac{i}{2\pi q} + O(q^{-2})$ .

## 5. THE RENORMALIZATION PRINCIPLE

**5.1. Heuristic.** For  $\alpha$  near 0, the near-parabolic renormalization  $\mathcal{R}$  of Inou and Shishikura [8] assigns to  $f_{e^{2\pi i \alpha}}$  a germ  $\mathcal{R}f_{e^{2\pi i \alpha}}$  at 0 with multiplier  $e^{-2\pi i/\alpha}$ . By parabolic implosion [14, 19], as  $\alpha \rightarrow 0$  with  $-1/\alpha \rightarrow \sigma \pmod{\mathbb{Z}}$ , the germ  $\mathcal{R}f_{e^{2\pi i \alpha}}$  tends to  $e^{2\pi i \sigma}$  times the horn germ. For  $\alpha = 1/q$  the phase is 0, and  $\mathcal{R}f_{e^{2\pi i 1/q}}$  is a parabolic germ with multiplier one. The rotation quotient  $Q_{1/q}$  of Corollary 3.2 is also a germ with multiplier one, and its 3-jet, which determines  $\kappa(1/q)$ , is determined by  $f^q$  (Remark 3.3). If the 3-jets of  $Q_{1/q}$  and of  $\mathcal{R}f_{e^{2\pi i 1/q}}$  are formally conjugate, then continuity of  $\mathcal{R}$  gives the first statement of Conjecture 4.5.

A proof along these lines needs two steps that we have not carried out: (a) the identification of  $\iota(Q_{1/q})$  with the index of  $\mathcal{R}f_{e^{2\pi i 1/q}}$ , that is, of the parabolic flower of  $f^q$  at 0 with the Fatou coordinates used to define  $\mathcal{R}$ ; and (b) the convergence  $\mathcal{R}f_{e^{2\pi i 1/q}} \rightarrow \mathcal{P}_0$  uniformly on a fixed disc, which should follow from parabolic implosion and gives convergence of the index by the residue formula. The rate  $O(q^{-2})$  in Conjecture 4.5 requires more, namely the first-order statement of Conjecture 5.2 below. The same reasoning applies to the whole unfolding, and there the phase can be computed exactly.

**Proposition 5.1** (Kinematics). *Let  $\lambda = e^{2\pi i \alpha} = \lambda_0 e^{u/q^2}$  with  $\lambda_0 = e^{2\pi i p/q}$ . Then  $e^{-2\pi i/\alpha} = \mu e^{\tilde{u}/p^2}$ , where  $\mu = e^{-2\pi i q/p}$  and*

$$\tilde{u} = \frac{u}{1 + u/(2\pi i p q)} = u + \frac{i u^2}{2\pi p q} + O\left(\frac{u^3}{q^2}\right).$$

*Proof.*  $\alpha = (2\pi i p q + u)/(2\pi i q^2)$ , so  $-1/\alpha + q/p = q u / (p(2\pi i p q + u))$ , and  $2\pi i p^2$  times this is  $\tilde{u}$ .  $\square$

Fix  $p \geq 1$  and  $r$  with  $\gcd(p, r) = 1$ , and let  $q = pN + r$ . Then  $\mu = e^{-2\pi i r/p}$  does not depend on  $N$ . Let  $\rho_{p,r}(u)$  be the multiplier of the  $p$ -cycle of

$$\mathcal{M}_u(W) = \mu e^{u/p^2} \mathcal{P}_0(W)$$

that bifurcates from  $W = 0$ , and write  $\rho_{p,r}(u) = 1 - u + \kappa_{p,r} u^2 + \rho_3 u^3 + \dots$ . By Theorem 2.1 and Remark 2.2, applied to the family  $\mu e^\varepsilon \mathcal{P}_0$  with  $\varepsilon = u/p^2$ , the first two coefficients are 1 and  $-1$  provided  $(\mu \mathcal{P}_0)^p(W) = W + aW^{p+1} + \dots$  with  $a \neq 0$ ; we verified  $a \neq 0$  numerically in all classes considered. Let  $u_a$  be the endpoint of the arc  $\rho_{p,r}^{-1}([-1, 1])$  that issues from  $u = 0$ .

**Conjecture 5.2** (Renormalization principle). For fixed  $(p, r)$  and  $q = pN + r \rightarrow \infty$ ,

$$R_{p/q}(u) = \rho_{p,r}(\tilde{u}(u)) + O(q^{-2})$$

coefficientwise, and uniformly in  $u$  on a neighbourhood of the arc from 0 to  $u_a$ .

Comparing coefficients gives the following consequences.

**Corollary 5.3** (conditional). *Under Conjecture 5.2,*

$$(5) \quad \kappa(p/q) = \kappa_{p,r} - \frac{i}{2\pi p q} + O(q^{-2}), \quad r_3(p/q) = \rho_3 + \frac{i \kappa_{p,r}}{\pi p q} + O(q^{-2}),$$

where, by Theorem 2.1 applied to  $\mu e^\varepsilon \mathcal{P}_0$ ,

$$(6) \quad \kappa_{p,r} = \frac{\iota((\mu \mathcal{P}_0)^p) - \frac{1}{2}}{p}.$$

Moreover  $G(p/q) \rightarrow \frac{1}{2}|u_a|$ , by Hurwitz's theorem and Definition 2.4, provided  $\rho'_{p,r}(u_a) \neq 0$ .

In particular the first-order term  $-i/(2\pi p q)$  involves no property of the germ, which explains why it is the same in degree three (Section 8). We now test the three parts of Conjecture 5.2: the limits of  $\kappa$ , the whole unfolding, and the first-order correction.

**5.2. Bounded  $p$ .** Table 3 compares (6) with bulb data for all nine classes with  $2 \leq p \leq 5$ . The bulb values are normal-form values of  $\kappa(p/q)$  for  $64 \leq N \leq 1024$ , extrapolated in  $1/q$ . All nine agree to between  $6 \cdot 10^{-9}$  and  $2.6 \cdot 10^{-8}$ . This is within the extrapolation uncertainty, which we take to be the spread of stable fits and which reaches  $2.5 \cdot 10^{-7}$ . The opposite phase convention,  $\mu = e^{+2\pi i r/p}$ , exchanges the classes  $r$  and  $p - r$  and fails. Figure 2 places these limits within the set of all  $\kappa(p/q)$  at a fixed  $q$ .

| $p$ | $r$ | horn-map prediction $(\iota((\mu\mathcal{P}_0)^p) - \frac{1}{2})/p$ | bulbs, extrapolated                 | $ \Delta $          |
|-----|-----|---------------------------------------------------------------------|-------------------------------------|---------------------|
| 2   | 1   | $0.049693182903 - 0.023894096844i$                                  | $0.049693177608 - 0.023894071809i$  | $2.6 \cdot 10^{-8}$ |
| 3   | 1   | $0.054529447219 - 0.0050171894089i$                                 | $0.054529443083 - 0.0050171743067i$ | $1.6 \cdot 10^{-8}$ |
| 3   | 2   | $0.055036036944 - 0.023467983453i$                                  | $0.055036033035 - 0.023467969651i$  | $1.4 \cdot 10^{-8}$ |
| 4   | 1   | $0.053737282786 + 0.0074895906470i$                                 | $0.053737279652 + 0.0074896009583i$ | $1.1 \cdot 10^{-8}$ |
| 4   | 3   | $0.054739672309 - 0.026348641547i$                                  | $0.054739669177 - 0.026348633046i$  | $9.1 \cdot 10^{-9}$ |
| 5   | 1   | $0.051130771685 + 0.015984969447i$                                  | $0.051130769387 + 0.015984976959i$  | $7.9 \cdot 10^{-9}$ |
| 5   | 2   | $0.059925460063 - 0.010450443803i$                                  | $0.059925457760 - 0.010450436333i$  | $7.8 \cdot 10^{-9}$ |
| 5   | 3   | $0.060335706660 - 0.0045365352268i$                                 | $0.060335704038 - 0.0045365278901i$ | $7.8 \cdot 10^{-9}$ |
| 5   | 4   | $0.052624850350 - 0.029174853238i$                                  | $0.052624847638 - 0.029174847754i$  | $6.1 \cdot 10^{-9}$ |

TABLE 3. Bounded- $p$  limits of  $\kappa$  (degree two): the prediction (6) against the bulb values extrapolated from  $q = pN + r$ ,  $64 \leq N \leq 1024$ .

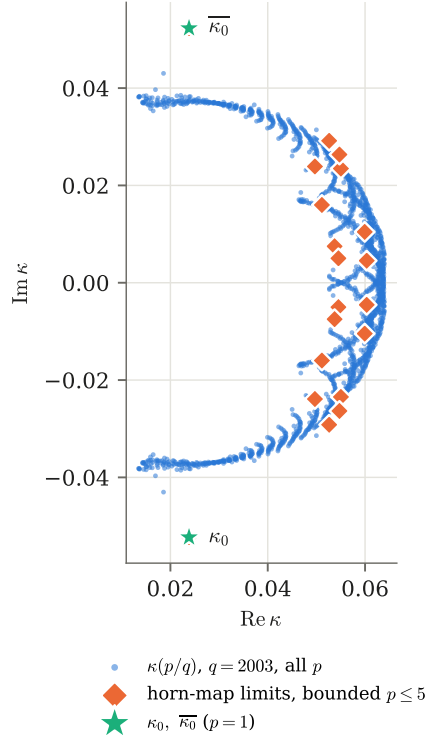


FIGURE 2. All  $\kappa(p/q)$  for  $q = 2003$  (values for  $p > q/2$  from the symmetry  $\kappa(1 - p/q) = \overline{\kappa(p/q)}$ ), the limits (6) for  $p \leq 5$ , and  $\kappa_0, \overline{\kappa_0}$ . The  $p = 1$  bulbs have the extreme values of  $\text{Im } \kappa$ .

**5.3. The unfolding and the bulb size.** For  $p = 1$  the Taylor coefficients  $\rho_k$  of  $\rho_{1,1}$  agree with the coefficients  $r_k(1/1025)$  through order eight up to the finite- $q$  shift; for instance  $\rho_3 = -3.78 \cdot 10^{-4} - 9.80 \cdot 10^{-4}i$  against  $r_3(1/1025) = -3.62 \cdot 10^{-4} - 9.72 \cdot 10^{-4}i$ , and  $\rho_8 = 3.14 \cdot 10^{-9} - 4.56 \cdot 10^{-9}i$  against  $r_8(1/1025) = 3.13 \cdot 10^{-9} - 4.50 \cdot 10^{-9}i$ . Since  $r_3$  depends on the unfolding and not only on the germ, this checks that  $u$  enters as the Lavaurs phase. The shift  $r_3 - \rho_3 = 1.6 \cdot 10^{-5} + 0.8 \cdot 10^{-5}i$  is the one predicted by (5), namely  $1.62 \cdot 10^{-5} + 0.74 \cdot 10^{-5}i$ . Table 4 compares the predicted bulb sizes with the bulbs for all ten classes with  $p \leq 5$ . The discrepancies are at most  $9 \cdot 10^{-7}$ .

The bulb values are extrapolated from three values of  $N$  with three parameters, and some sequences are not monotone at the level of  $10^{-7}$ , so we regard  $10^{-6}$  as their uncertainty. For  $p = 1$  this gives

$$\lim_{q \rightarrow \infty} q^3 \delta(1/q) = 2\pi G_0, \quad G_0 = 1.02641539 \dots,$$

so the  $1/q$ -bulbs, which accumulate at the cusp  $c = \frac{1}{4}$ , have diameters  $2\pi G_0 q^{-3}(1 + O(q^{-1}))$ .

**5.4. The first-order correction.** The first identity in (5) is the coefficient  $\gamma_1 = -i/(2\pi)$  of Numerical result 4.1. For  $2 \leq p \leq 5$  the fitted first-order coefficients satisfy  $\gamma_1(p) \cdot 2\pi p/(-i) = 1.0008, 1.0011, 1.0014, 1.0016$ ; these fits are much less constrained than for  $p = 1$ , and we give no error bars for them. The following test uses all Taylor coefficients at once and involves no fitting.

**Numerical result 5.4.** Let  $r_k(p/q)$  be the Taylor coefficients of  $R_{p/q}$  and  $\rho_k$  those of  $\rho_{p,r}$ , both computed by Cauchy integrals on  $|u| = 1.5$ . For  $2 \leq k \leq 8$ , for  $(p, r) = (1, 1), (2, 1), (3, 1)$  and  $(3, 2)$ , and for  $N = 64, 128, \dots, 1024$ , the residuals  $|r_k(p/q) - [u^k]\rho_{p,r}(\tilde{u})|$  decrease by a factor between 3.77 and 4.28 per doubling of  $N$ . This holds in all 41 cases in which the residual exceeds  $10^{-8}$ ; below that level the round-off of the Cauchy integrals takes over. Without the substitution  $u \mapsto \tilde{u}$  the factor lies between 1.1 and 2.0. In degree three, with the cubic horn germ, Cauchy integrals on  $|u| = 1$  and  $p = 1$ , the factor lies between 3.75 and 4.06 (26 cases). Figure 3 shows the case  $p = 1$ .

Thus, numerically and through first order in  $1/q$ , the whole function  $R_{p/q}$ , and not only  $\kappa$ , is the renormalized family composed with the exact Inou–Shishikura phase; within this evidence, the renormalized germ itself has no  $O(1/q)$  correction.

## 6. THE HIERARCHY AND THE CONSTANT $\kappa_*$

**6.1. The iteration.** For  $\alpha = [0; m_1, m_2, \dots]$  the renormalization  $\mathcal{R}f_{e^{2\pi i\alpha}}$  has multiplier  $e^{-2\pi i/\alpha}$ . Conjugating by  $z \mapsto \bar{z}$  turns its rotation number into  $1/\alpha \bmod 1 = [0; m_2, m_3, \dots]$ , so that  $\mathcal{R}$  acts on continued fractions as the Gauss map [8]. When the first  $k$  partial quotients tend to infinity, each renormalization should therefore be close to  $\mathcal{P}$  applied to the complex conjugate of the previous one. Write  $\bar{\mathcal{G}}(W) = \overline{\mathcal{G}(\bar{W})}$ , normalize germs to the form  $v + v^2 + O(v^3)$ , and set

$$\mathcal{G}_1 = \mathcal{P}(f_1), \quad \mathcal{G}_{k+1} = \mathcal{P}(\bar{\mathcal{G}}_k), \quad \kappa_k = \iota(\mathcal{G}_k) - \frac{1}{2}.$$

By Lemma 4.3,  $\kappa_k = a_2/(2\pi i a_1^2)$  for the horn map of  $\bar{\mathcal{G}}_{k-1}$ , and  $\kappa_1 = \kappa_0$ . For  $k \geq 2$  the germs  $\mathcal{G}_k$  are given by convergent series of small radius (about 2.6 in the normalized coordinate  $v$  at  $k = 2$ ), and the horn maps must be sampled at heights where the orbits stay well inside the disc of convergence. Table 5 lists  $\kappa_1, \dots, \kappa_7$  for two independent settings of height, Fourier truncation and precision. They agree to  $2 \cdot 10^{-14}$  or better at every level.

**Numerical result 6.1.** The differences  $\Delta_k = \kappa_k - \kappa_{k-1}$  decrease geometrically:  $|\Delta_{k+1}|/|\Delta_k|$  tends to 0.0419, and the ratios are  $\Delta_{k+1}/\Delta_k \approx -0.01546 \mp 0.03894i$ , with imaginary parts of alternating sign (Figure 4). This is the signature of an antilinear contraction  $\Delta \mapsto \nu\bar{\Delta}$ , as expected since the linearization of  $\mathcal{G} \mapsto \mathcal{P}(\bar{\mathcal{G}})$  is antilinear. Fitting  $\Delta_{k+1} = L\Delta_k$  with  $L$  real-linear on  $\mathbb{C} \cong \mathbb{R}^2$  gives eigenvalues  $\pm 0.041899$  and the limit

$$\kappa_* := \lim_{k \rightarrow \infty} \kappa_k = 0.011929183413 - 0.037455493752i,$$

with the fits through levels 6 and 7 differing by less than  $2 \cdot 10^{-12}$ . The first Fourier coefficient  $a_1$  converges in the same way, to  $0.0409914 - 0.0089033i$ , which is consistent with convergence of  $\mathcal{G}_k$

| $p$ | $r$ | $G$ at $q = 512p + r$ | $G$ at $q = 2048p + r$ | extrapolated | horn map   |
|-----|-----|-----------------------|------------------------|--------------|------------|
| 1   | 1   | 1.02634395            | 1.02639633             | 1.02641535   | 1.02641539 |
| 2   | 1   | 1.11142282            | 1.11143038             | 1.11143322   | 1.11143318 |
| 3   | 1   | 1.12836657            | 1.12836757             | 1.12836781   | 1.12836776 |
| 3   | 2   | 1.12537076            | 1.12537450             | 1.12537590   | 1.12537582 |
| 4   | 1   | 1.12357397            | 1.12357381             | 1.12357368   | 1.12357348 |
| 4   | 3   | 1.12156857            | 1.12157111             | 1.12157204   | 1.12157195 |
| 5   | 1   | 1.11300416            | 1.11300362             | 1.11300313   | 1.11300337 |
| 5   | 2   | 1.14414695            | 1.14414738             | 1.14414703   | 1.14414795 |
| 5   | 3   | 1.14644523            | 1.14644563             | 1.14644539   | 1.14644595 |
| 5   | 4   | 1.11312789            | 1.11312962             | 1.11312984   | 1.11313044 |

TABLE 4. Bulb size  $G$  (Definition 2.4) along  $q = pN + r$ , extrapolated from  $N = 512, 1024, 2048$ , against the prediction of Corollary 5.3. The first row gives  $G_0 = \lim G(1/q)$ .

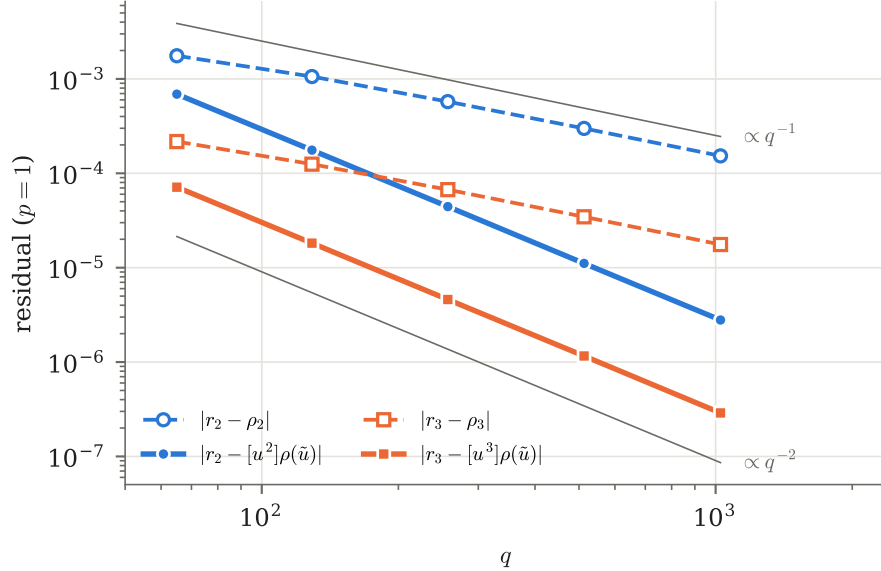


FIGURE 3. Residuals of the Taylor coefficients  $r_2, r_3$  of  $R_{1/q}$  against the renormalized family, with (solid) and without (dashed) the Möbius substitution of Proposition 5.1. The substitution turns the  $O(q^{-1})$  residual into an  $O(q^{-2})$  one.

to a fixed point  $\mathcal{G}_* = \mathcal{P}(\bar{\mathcal{G}}_*)$ . If  $\mathcal{G}_*$  exists, its cubic coefficient is  $\kappa_* + \frac{1}{2} = 0.5119292 - 0.0374555i$ ; we have not proved that it exists.

*Remark 6.2* (The holomorphic operator). Without the conjugation,  $\mathcal{G} \mapsto \mathcal{P}(\mathcal{G})$  is the parabolic renormalization operator whose fixed point  $f_*$  was shown to exist by Inou and Shishikura and computed by Lanford and Yampolsky [13]. They report  $f_*(z) \approx z + z^2 + (0.514 - 0.0346i)z^3$  and a leading eigenvalue  $\approx -0.017 + 0.040i$  of  $D\mathcal{P}$  at  $f_*$ . Our code without conjugation reproduces both: the cubic coefficients converge to  $0.5142768 - 0.0346276i$  (extrapolated,  $\pm 10^{-7}$ ), and the ratio of successive differences tends to  $-0.01764 + 0.04038i$ . The bulb data at the levels

| $k$      | $\kappa_k$ ( $h = 1$ )                | $ \kappa_k - \kappa_{k-1} $ | $ \kappa_k^{(h=1)} - \kappa_k^{(h=5/4)} $ |
|----------|---------------------------------------|-----------------------------|-------------------------------------------|
| 1        | 0.0238258874022 - 0.0523046591140 $i$ |                             | —                                         |
| 2        | 0.0112359215880 - 0.0377814219023 $i$ | $1.9 \cdot 10^{-2}$         | $6.1 \cdot 10^{-16}$                      |
| 3        | 0.0119510972606 - 0.0374792854625 $i$ | $7.8 \cdot 10^{-4}$         | $1.7 \cdot 10^{-14}$                      |
| 4        | 0.0119279615825 - 0.0374560782080 $i$ | $3.3 \cdot 10^{-5}$         | $5.8 \cdot 10^{-16}$                      |
| 5        | 0.0119292218902 - 0.0374555354669 $i$ | $1.4 \cdot 10^{-6}$         | $2.5 \cdot 10^{-17}$                      |
| 6        | 0.0119291812682 - 0.0374554947782 $i$ | $5.7 \cdot 10^{-8}$         | $9.9 \cdot 10^{-19}$                      |
| 7        | 0.0119291834806 - 0.0374554938253 $i$ | $2.4 \cdot 10^{-9}$         | $1.2 \cdot 10^{-19}$                      |
| $\infty$ | 0.011929183413 - 0.037455493752 $i$   |                             |                                           |

TABLE 5. The hierarchy  $\kappa_k = \iota(\mathcal{G}_k) - \frac{1}{2}$ ,  $\mathcal{G}_{k+1} = \mathcal{P}(\bar{\mathcal{G}}_k)$ , computed at heights  $h = 1$  (16 Fourier modes, 90 digits) and  $h = \frac{5}{4}$  (14 modes, 100 digits). Level 1 is  $\kappa_0$  and is common to both. The last row is the extrapolation of Numerical result 6.1.

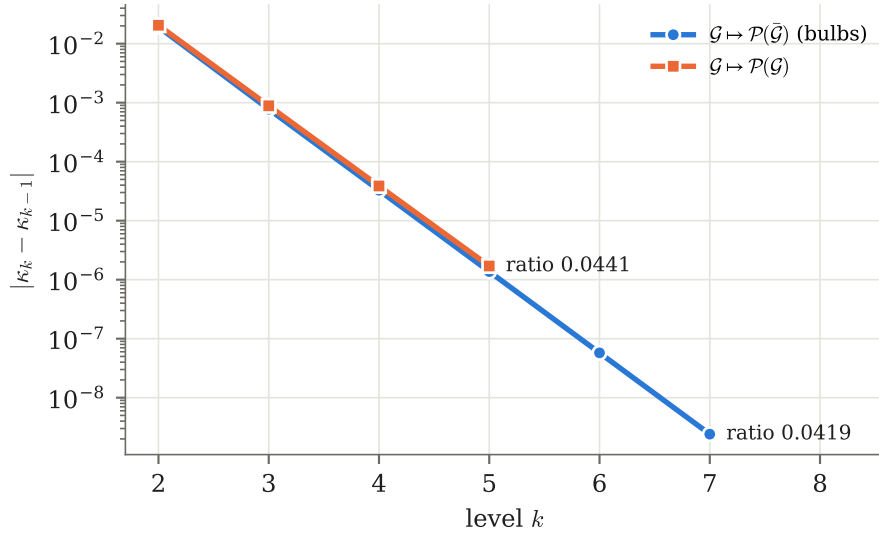


FIGURE 4. Successive differences of the two hierarchies:  $\mathcal{G} \mapsto \mathcal{P}(\bar{\mathcal{G}})$ , which governs the bulbs, and the holomorphic parabolic renormalization  $\mathcal{G} \mapsto \mathcal{P}(\mathcal{G})$  of Remark 6.2. The labels give the last ratio of successive differences.

$k \leq 3$  follow the conjugated operator instead, whose limit  $\kappa_*$  differs from the cubic coefficient of  $f_*$  minus  $\frac{1}{2}$ .

## 6.2. Comparison with bulbs.

**Conjecture 6.3.** For every  $k \geq 1$ ,

$$\lim_{N_k \rightarrow \infty} \cdots \lim_{N_1 \rightarrow \infty} \kappa([0; N_1, \dots, N_k]) = \begin{cases} \kappa_k, & k \text{ odd,} \\ \overline{\kappa_k}, & k \text{ even.} \end{cases}$$

In particular  $\kappa_*$  and  $\overline{\kappa_*}$  are the limits of the first correction as the number of large partial quotients of  $p/q$  grows, along odd and even  $k$  respectively.

The case  $k = 1$  is Numerical result 4.4. For  $k = 2$ , formula (6) with  $r = 1$  gives the limit of  $\kappa([0; N, p])$  as  $N \rightarrow \infty$  as an index of  $(e^{-2\pi i/p} \mathcal{P}_0)^p$ . We computed this index for  $p \leq 256$  with the normal form of Theorem 3.1 generalized to arbitrary germs, an  $O(p^3)$  recursion, and extrapolated in  $1/p$ . The result,  $0.0112359330 + 0.0377814233i$ , equals  $\overline{\kappa_2}$  to  $1.2 \cdot 10^{-8}$  on the best fit; fits of lower order differ from it by up to  $8 \cdot 10^{-7}$ . The conjugate appears because  $p^{-1} \bmod q \rightarrow 0^-$  along this sequence, against  $0^+$  for  $p = 1$ . Without the conjugation in the definition of  $\mathcal{G}_2$  one obtains  $0.014770 - 0.033911i$ , which the data exclude.

We also tested the diagonal sequences  $[0; N, \dots, N]$ , computing  $\kappa$  with the normal form at 128 and 192 bits (the two agree to  $4 \cdot 10^{-35}$ ) and fitting polynomials in  $1/N$ . For  $k = 2$  and  $16 \leq N \leq 160$  ( $q \leq 25601$ ) the limit agrees with  $\overline{\kappa_2}$  to  $3 \cdot 10^{-10}$ , so the diagonal and the iterated limits coincide. For  $k = 3$  and  $6 \leq N \leq 40$  ( $q \leq 64080$ ) the best fit ( $N \geq 14$ , degree  $J = 6$ ) gives  $0.0119509937 - 0.0374804294i$ , which agrees with  $\kappa_3$  to  $1.1 \cdot 10^{-6}$ . The fits of degree  $J = 3, 4, 5, 6$  on  $N \geq 14$  lie at distances  $8 \cdot 10^{-5}$ ,  $2.6 \cdot 10^{-5}$ ,  $4.5 \cdot 10^{-6}$  and  $1.1 \cdot 10^{-6}$  from  $\kappa_3$ , so we take  $10^{-5}$  as the uncertainty. For comparison,  $|\kappa_3 - \kappa_2| = 7.8 \cdot 10^{-4}$  and  $|\kappa_3 - \overline{\kappa_3}| = 7.5 \cdot 10^{-2}$ . Thus Conjecture 6.3, including its parity rule, holds numerically for  $k \leq 3$ . Level 4, where  $|\kappa_4 - \kappa_3| = 3.3 \cdot 10^{-5}$ , would require periods  $q \approx N^4$  with  $N \gtrsim 30$ .

## 7. LIMITS AT THE ROOTS OF SATELLITES

**7.1. One-sided limits at rationals.** Let  $p/q = [0; a_1, \dots, a_n]$  be in lowest terms, and let  $t$  be a finite tail of partial quotients ( $t = \emptyset$  allowed). As  $N \rightarrow \infty$  the satellites  $[0; a_1, \dots, a_n, N, t]$  of the cardioid accumulate at the root  $c(e^{2\pi i p/q})$  of  $W_{p/q}$ : from above if  $n$  is even and from below if  $n$  is odd, in terms of the internal angle. For  $p/q = 0/1$  ( $n = 0$ ) these are the bounded- $p$  sequences of Section 5, since  $[0; N, t] = p_t/(p_t N + r_t)$  with  $r_t/p_t = [0; t]$  (and  $p_t = r_t = 1$  for  $t = \emptyset$ ). The renormalization picture suggests that at the root of  $W_{p/q}$  the cusp germ  $z + z^2$  is replaced by  $f_{p/q}(z) := e^{2\pi i p/q} z + z^2$ , a parabolic germ with  $q$  petals.

For  $g = f_{p/q}^q = z + az^{q+1} + O(z^{q+2})$  the argument of Lemma 7.4 below applies verbatim: there is a unique formal Fatou coordinate

$$\Phi(z) = -\frac{1}{qa} z^{-q} + \sum_{\substack{-q < k \leq K \\ k \neq 0}} s_k z^k + \beta \log z + \dots, \quad \Phi \circ g = \Phi + 1,$$

without constant term, since the coefficient of  $z^{k+q}$  in  $\Phi \circ g - \Phi$  contains  $s_k$  with the factor  $ka$  and that of  $z^q$  contains  $\beta$  with the factor  $a$ . Its logarithmic coefficient is  $\beta = \text{résit}(g) = \frac{q+1}{2} - \iota_{p/q}$  (we checked this numerically for  $p/q = 1/3, 1/4$  and  $2/5$ ). Let  $P_0$  be an attracting petal of  $g$  and  $R_0$  the repelling petal that follows it counterclockwise. The  $g$ -orbits of points of  $R_0$  at height  $\text{Im } Z$  in the repelling Fatou coordinate reach the attracting petal that follows  $R_0$  counterclockwise when  $\text{Im } Z$  is above a bounded band, and  $P_0$  when it is below; within the band they may escape. We determine the band numerically, and it is not centred at the naive value, because the terms  $z^{-k}$ ,  $k < q$ , of  $\Phi$  have imaginary parts on  $R_0$  that depend on the depth. An orbit that lands in the  $j$ -th petal is carried back to  $P_0$  by  $f_{p/q}^m$  with  $m \equiv -jp^{-1} \pmod{q}$ , which is a step of  $m/q$  in Fatou coordinates. This defines an upper horn map  $E_{p/q}^+(Z) = Z + a_0 + \sum_{n \geq 1} a_n e^{2\pi i n Z}$  ( $\text{Im } Z \rightarrow +\infty$ ) and a lower horn map  $E_{p/q}^-(Z) = Z + b_0 + \sum_{n \geq 1} b_n e^{-2\pi i n Z}$  ( $\text{Im } Z \rightarrow -\infty$ ), with parabolic germs  $\mathcal{P}_{p/q}^+(W) = W \exp(2\pi i \sum a_n W^n)$  in  $W = e^{2\pi i Z}$  and  $\mathcal{P}_{p/q}^-(W) = W \exp(-2\pi i \sum b_n W^n)$  in  $W = e^{-2\pi i Z}$ . By Lemma 4.3,

$$\iota(\mathcal{P}_{p/q}^+) - \frac{1}{2} = \frac{a_2}{2\pi i a_1^2}, \quad \iota(\mathcal{P}_{p/q}^-) - \frac{1}{2} = -\frac{b_2}{2\pi i b_1^2}.$$

For  $p/q = 0/1$  this is  $\mathcal{P}_0$  and  $\kappa_0$  (and its conjugate for the lower map). For  $q \leq 2$  the germ  $f_{p/q}$  is real, and the two maps are complex conjugate; for  $q \geq 3$  they are unrelated.

**Conjecture 7.1** (Root limits). With  $\mathcal{P}_{p/q}^+$  if  $n$  is even and  $\mathcal{P}_{p/q}^-$  if  $n$  is odd, and  $\mu_t = e^{-2\pi i r_t/p_t}$ ,

$$\lim_{N \rightarrow \infty} \kappa([0; a_1, \dots, a_n, N, t]) = \frac{\iota((\mu_t \mathcal{P}_{p/q}^\pm)^{p_t}) - \frac{1}{2}}{p_t}.$$

In particular,  $\kappa$  has at every rational  $p/q$  the one-sided limits  $\iota(\mathcal{P}_{p/q}^\pm) - \frac{1}{2}$  along  $t = \emptyset$ .

For  $p/q = 0/1$  this is Conjecture 4.5 together with (6).

**Numerical result 7.2.** Table 6 compares Conjecture 7.1 with the bulbs for  $p/q = 1/3, 1/4, 2/5, 3/7$  from both sides, and at  $1/3$  also for the tails  $t = 2$  (both sides) and  $t = 3$ . The bulb values are normal-form values for  $N = 128, \dots, 2048$ , extrapolated by polynomials of degree four in  $1/q$ ; changing the degree to three moves them by at most  $1.3 \cdot 10^{-9}$ . All eleven limits agree with the horn maps to within  $6 \cdot 10^{-11}$ .

For  $q \geq 3$  the two one-sided limits at  $p/q$  are not complex conjugates; at  $1/3$  they are  $0.0158408 + 0.0401752i$  from below and  $0.0165812 - 0.0395047i$  from above. The symmetry  $\kappa(1-x) = \overline{\kappa(x)}$  instead relates  $p/q$  from above to  $(q-p)/q$  from below. The first correction therefore carries, at every rational, a pair of constants of the same kind as  $\kappa_0$ . They are invariants of the parabolic germ  $f_{p/q}$  at the root of  $W_{p/q}$ .

**7.2. The period-two disc.** Conjecture 7.1 concerns the satellites of the cardioid near the root of  $W_{p/q}$ . The satellites of  $W_{p/q}$  itself accumulate at the same point, and Remark 2.2 applies to them. For the period-two disc  $W_{1/2}$ ,  $c = -1 + \lambda/4$ , with  $\lambda$  the multiplier of the attracting 2-cycle, we write  $\kappa_{W_{1/2}}(p/q)$  for the coefficient of (1). At generic  $p/q$  it is close to the value of the cardioid at the same  $p/q$ : for  $p/q = [0; 2, 9, 4, 256]$ ,  $[0; 3, 5, 256]$  and  $[0; 5, 1, 2, 256]$  the two differ by  $3.8 \cdot 10^{-6}$ ,  $3.7 \cdot 10^{-5}$  and  $7.8 \cdot 10^{-5}$ . The bounded- $p$  limits, however, differ from those of the cardioid; for instance  $\lim_{q \rightarrow \infty} \kappa_{W_{1/2}}(1/q) \neq \kappa_0$ . The following conjecture attributes them to the germ at the root.

**Conjecture 7.3** (Root principle). The limits of  $\kappa$  and of  $G$  along satellites  $[0; N, t]$  of a hyperbolic component, as  $N \rightarrow \infty$  with the tail  $t$  fixed, depend only on the parabolic germ at the root of the component. In particular the satellite  $[0; N, t]$  of  $W_{1/2}$  and the satellite  $[0; 1, 1, N-1, t]$  of the cardioid, which is governed by  $\mathcal{P}_{1/2}^+$  (Conjecture 7.1), have the same limits.

Here  $[0; N, t] \mapsto [0; 1, 1, N-1, t]$  is the map  $x \mapsto 1/(2-x)$ . Table 7 confirms the second statement for three tails, and for the bulb size. The best fits agree to about  $10^{-9}$ ; changing the degree of the fit from two to three moves the limits by at most  $1.3 \cdot 10^{-7}$ , which we take as the uncertainty. The disc values come from Cauchy integrals in the multiplier of the 2-cycle, and the cardioid values from the normal form. When the first partial quotient stays bounded the correspondence fails: at  $N = 256$  the two sides differ by between  $3 \cdot 10^{-5}$  and  $1.5 \cdot 10^{-3}$  for the prefixes  $(2, 9, 4)$ ,  $(3, 5)$ ,  $(1, 4)$ ,  $(5, 1, 2)$  and  $(7)$ , and the disc is then closer to the cardioid at the same  $p/q$ . This is consistent with the picture in which only the first renormalization, and hence the germ at the root, distinguishes components.

**7.3. The  $\frac{1}{2}$ -root in exact arithmetic.** At  $c = -\frac{3}{4}$  the map is conjugate to  $f(z) = -z + z^2$ , and  $g = f \circ f = z - 2z^3 + z^4$  has a fixed point of multiplicity three at 0. Its attracting petals lie along  $\pm\mathbb{R}$  and its repelling petals along  $\pm i\mathbb{R}$ , and  $f$  exchanges the members of each pair. In this case the Fatou coordinate has rational coefficients.

| $p/q$ | side  | $t$         | horn map of $e^{2\pi ip/q}z + z^2$   | bulbs, extrapolated                  | $ \Delta $           |
|-------|-------|-------------|--------------------------------------|--------------------------------------|----------------------|
| 1/3   | below | $\emptyset$ | $0.015840773234 + 0.040175221242 i$  | $0.015840773241 + 0.040175221243 i$  | $6.5 \cdot 10^{-12}$ |
| 1/3   | above | $\emptyset$ | $0.016581162594 - 0.039504714589 i$  | $0.016581162601 - 0.039504714589 i$  | $6.4 \cdot 10^{-12}$ |
| 1/4   | below | $\emptyset$ | $0.014455842876 + 0.039068895117 i$  | $0.014455842881 + 0.039068895118 i$  | $5.3 \cdot 10^{-12}$ |
| 1/4   | above | $\emptyset$ | $0.015760467204 - 0.037801851383 i$  | $0.015760467209 - 0.037801851382 i$  | $5.3 \cdot 10^{-12}$ |
| 2/5   | below | $\emptyset$ | $0.014434584508 + 0.037505178012 i$  | $0.014434584513 + 0.037505178012 i$  | $4.6 \cdot 10^{-12}$ |
| 2/5   | above | $\emptyset$ | $0.014296912442 - 0.037686119109 i$  | $0.014296912446 - 0.037686119110 i$  | $4.6 \cdot 10^{-12}$ |
| 3/7   | above | $\emptyset$ | $0.013301284920 - 0.037190044710 i$  | $0.013301284861 - 0.037190044712 i$  | $5.9 \cdot 10^{-11}$ |
| 3/7   | below | $\emptyset$ | $0.013998687363 + 0.036367863348 i$  | $0.013998687363 + 0.036367863334 i$  | $1.4 \cdot 10^{-11}$ |
| 1/3   | above | 2           | $0.046849904480 - 0.018242741860 i$  | $0.046849904482 - 0.018242741860 i$  | $2.7 \cdot 10^{-12}$ |
| 1/3   | below | 2           | $0.046510029333 + 0.018510459943 i$  | $0.046510029336 + 0.018510459943 i$  | $2.8 \cdot 10^{-12}$ |
| 1/3   | above | 3           | $0.052913142048 - 0.0017286435024 i$ | $0.052913142050 - 0.0017286435020 i$ | $1.7 \cdot 10^{-12}$ |

TABLE 6. One-sided limits of  $\kappa$  at rationals: the upper (“above”) or lower (“below”) horn map of  $e^{2\pi ip/q}z + z^2$ , with the tail rule of Conjecture 7.1, against the bulbs  $[0; a_1, \dots, a_n, N, t]$  extrapolated from  $N = 128, \dots, 2048$ . The row 2/5 below equals the complex conjugate of 3/5 above, by the symmetry  $\kappa(1 - x) = \overline{\kappa(x)}$ .

|                         | disc $[0; N, t]$                   | cardioid $[0; 1, 1, N - 1, t]$     | $ \Delta $           |
|-------------------------|------------------------------------|------------------------------------|----------------------|
| $\kappa, t = \emptyset$ | $0.01840526173 - 0.04290372357 i$  | $0.01840526170 - 0.04290372213 i$  | $1.4 \cdot 10^{-9}$  |
| $\kappa, t = 2$         | $0.04755255860 - 0.01974956083 i$  | $0.04755255811 - 0.01974956031 i$  | $7.1 \cdot 10^{-10}$ |
| $\kappa, t = 3$         | $0.05330701166 - 0.002605955494 i$ | $0.05330701184 - 0.002605955354 i$ | $2.2 \cdot 10^{-10}$ |
| $G, t = \emptyset$      | 1.020162587                        | 1.020162587                        | $5.3 \cdot 10^{-10}$ |

TABLE 7. The period-two disc near its root against the cardioid near its  $\frac{1}{2}$ -root. The disc column uses  $[0; N, t]$  in the multiplier of the 2-cycle, the cardioid column  $[0; 1, 1, N - 1, t]$ . Both are extrapolated from  $N = 128, \dots, 2048$  by polynomials of degree three in  $1/q$ .

**Lemma 7.4** (Two-petal Fatou coordinate). *There is a unique formal series*

$$\Phi(z) = \frac{1}{4}z^{-2} + s_{-1}z^{-1} + \beta \log z + \sum_{k \geq 1} s_k z^k$$

without constant term such that  $\Phi \circ g = \Phi + 1$ . Its coefficients are rational:  $s_{-1} = \frac{1}{4}$ ,  $\beta = \frac{11}{8}$ ,  $s_1 = -\frac{5}{16}$ ,  $s_2 = \frac{75}{64}$ ,  $\dots$ . The coefficient  $\beta$  is the *résidu itératif*  $\frac{3}{2} - \iota(g)$  of  $g$ , and  $\iota(g) = \iota_{1/2} = \frac{1}{8}$ .

*Proof.* Write  $g(z) = z(1 + h)$  with  $h = -2z^2 + z^3$ . Then  $\Phi \circ g - \Phi = \sum_k s_k z^k ((1 + h)^k - 1) + \beta \log(1 + h)$  (with  $s_{-2} = \frac{1}{4}$ ), and  $z^k((1 + h)^k - 1) = -2kz^{k+2} + O(z^{k+3})$ . The coefficient of  $z^m$  therefore contains  $s_{m-2}$  with the factor  $-2(m-2)$  for  $m \neq 2$ , and  $\beta$  with the factor  $-2$  for  $m = 2$ , together with unknowns of lower index. Solving order by order gives existence and uniqueness, the constant term being free, and the stated values. The index is  $\text{Res } dz/(2z^3 - z^4) = \frac{1}{8}$ , which agrees with  $\iota_{1/2}$  in Remark 2.3.  $\square$

In the construction of Section 7.1, orbits leaving the repelling petal  $+i\mathbb{R}$  above the escape band (which lies between the heights 1.5 and 3) reach the attracting petal  $-i\mathbb{R}$ , and  $f$ , a half step of  $g$ , carries them to  $+\mathbb{R}$ . We write  $\mathcal{P}_{1/2} = \mathcal{P}_{1/2}^+$  and  $\kappa_{1/2} := \iota(\mathcal{P}_{1/2}) - \frac{1}{2}$ .

**Numerical result 7.5.** Computed at the heights 4.5 and 5.5 with 100 digits, which agree in all digits shown,

$$\begin{aligned}\operatorname{Re} \kappa_{1/2} &= 0.018405261617964781008961965439681, \\ \operatorname{Im} \kappa_{1/2} &= -0.042903723880230475207353337798906.\end{aligned}$$

This agrees with the bulb limits of Table 7 within their uncertainty of about  $10^{-7}$ ; the best fits differ from it by  $3 \cdot 10^{-10}$  for the disc and  $2 \cdot 10^{-9}$  for the cardioid. The germ  $\mathcal{P}_{1/2}$  also carries the rest of the theory for the disc. With the phase  $e^{-2\pi ir/p}$  and  $r = 1$  (since  $[0; N, t] = t/(tN + 1)$ ), formula (6) gives  $0.04755255819 - 0.01974956104i$  for  $t = 2$  and  $0.05330701191 - 0.00260595580i$  for  $t = 3$ . The unfolding  $e^u \mathcal{P}_{1/2}$  gives  $G = 1.0201625875$  for  $p = 1$ , and  $-e^{u/4} \mathcal{P}_{1/2}$  gives  $G = 1.1070111364$  for  $p = 2$ , against 1.107011147 for disc bulbs along  $2/(2N + 1)$ .

The first-order coefficient of the disc,  $\kappa_{W_{1/2}}(1/q) = \kappa_{1/2} + (-0.17743 - 0.19381i)/q + O(q^{-2})$ , is not the kinematic value  $-i/(2\pi q)$  of Corollary 5.3. Proposition 5.1 concerns the multiplier whose rotation number is renormalized. At  $c = -\frac{3}{4}$  the natural candidate for that role is the multiplier  $\nu$  of the fixed point of  $f$ , not the multiplier  $\lambda = e^s$  of the 2-cycle, and  $\log(-\nu) = -\frac{s}{4} - \frac{11}{64}s^2 - \frac{51}{512}s^3 + \dots$ ; note that  $\frac{11}{64} = \beta/8$ . The kinematics of the disc therefore differ from those of the cardioid already at first order. Deriving the coefficient  $-0.17743 - 0.19381i$  from this is an open problem.

## 8. DEGREE THREE: A SECOND UNIVERSALITY CLASS

The main component of  $z^3 + c$  has two cusps,  $c = \pm 2/(3\sqrt{3})$ , exchanged by  $c \mapsto -c$ . Near  $c_* = 2/(3\sqrt{3})$  we parametrize it by the multiplier  $\lambda$  of the fixed point  $z = +(\lambda/3)^{1/2}$ , with  $c = z - z^3$ ; the map  $\lambda \mapsto c$  is two-to-one on the whole component, and we use this branch. At  $\lambda = 1$ ,  $z_* = 1/\sqrt{3}$ , and the parabolic germ is  $w \mapsto w + \sqrt{3}w^2 + w^3$ . In  $v = \sqrt{3}w$  this is

$$g(v) = v + v^2 + \frac{1}{3}v^3,$$

whose résidu itératif is  $1 - \frac{1}{3} = \frac{2}{3}$ . Lemma 4.2 generalizes with  $F(w) = -1/g(-1/w)$  and  $\Phi(w) = w - \frac{2}{3} \log w + \sum c_k w^{-k}$ , where  $c_1, c_2, \dots = \frac{2}{9}, \frac{5}{54}, \frac{14}{243}, \frac{89}{2430}, \dots$ . The horn map gives, at 50 and at 70 digits,

$$\kappa_0^{(3)} := \frac{a_2}{2\pi i a_1^2} = 0.143599278095291569488064104088 - 0.030311647898502537611943110858i.$$

Near-parabolic renormalization for unicritical maps of this degree was developed by Chéritat [10] and Yang [20], and the heuristics of Section 5 apply with the cusp germ. Write  $\kappa^{(3)}(p/q)$  for the coefficient of (1) in this family. Table 8 shows that  $\kappa^{(3)}(1/q) + \frac{i}{2\pi q}$  converges to  $\kappa_0^{(3)}$  like  $q^{-2}$ . The correction  $-i/(2\pi q)$  is thus the same as in degree two, as Corollary 5.3 predicts, while the constant is not. The bounded- $p$  predictions hold in degree three as well:  $\kappa$  agrees to within  $5 \cdot 10^{-6}$  for  $(p, r) = (1, 1), (2, 1), (3, 1), (3, 2)$  at  $q \approx 512p$ .

The bulb size needs one more observation.

The following statement is classical [16]; we include a short proof, since we use its last part.

**Proposition 8.1.** *Let  $W$  be a hyperbolic component of  $z^d + c$ , of period  $n$ . Its multiplier map  $\rho : W \rightarrow \mathbb{D}$  has the form  $\rho = \phi^{d-1}$  with  $\phi : W \rightarrow \mathbb{D}$  a conformal isomorphism. Consequently  $\rho^{-1}((-1, 0])$  consists of  $d - 1$  arcs issuing from the centre, and each of them converges to a single point of  $\partial W$  at which the cycle has multiplier  $-1$ .*

| $q$                                    | $\kappa^{(3)}(1/q)$              | $ \kappa^{(3)}(1/q) + \frac{i}{2\pi q} - \kappa_0^{(3)} $ |
|----------------------------------------|----------------------------------|-----------------------------------------------------------|
| 128                                    | $0.1436180525 - 0.03147252376 i$ | $8.5 \cdot 10^{-5}$                                       |
| 256                                    | $0.1436043106 - 0.03091278633 i$ | $2.1 \cdot 10^{-5}$                                       |
| 512                                    | $0.1436005784 - 0.03061736685 i$ | $5.3 \cdot 10^{-6}$                                       |
| 1024                                   | $0.1435996089 - 0.03046578967 i$ | $1.3 \cdot 10^{-6}$                                       |
| horn map of $v + v^2 + \frac{1}{3}v^3$ |                                  | $0.1435992781 - 0.03031164790 i$                          |

TABLE 8. Degree three:  $\kappa^{(3)}(1/q)$  from Cauchy integrals on  $|u| = 1$  (Appendix A) against the horn map of the cubic cusp germ.

*Proof.* The map  $\rho$  is proper, since the cycle becomes indifferent at  $\partial W$ . Its zeros are the parameters at which the critical point lies on the cycle, and there is exactly one, the centre  $c_0$ , because  $W$  is a simply connected domain with a unique centre [16]. Let  $P_n(c) = f_c^n(0)$ . Then  $P_1 = c$  and  $P'_{k+1} = dP_k^{d-1}P'_k + 1$ , so  $P'_n \equiv 1 \pmod{d}$  as a polynomial with integer coefficients, and  $P'_n(c_0) \neq 0$  because  $1/d$  is not an algebraic integer (Gleason's argument). Near  $c_0$  the cycle point  $z_0(c)$  close to 0 solves  $f_c^n(z) = z$ , and  $\partial_z(f_c^n(z) - z) = -1$  at  $(c_0, 0)$ ; by the implicit function theorem  $z_0$  is holomorphic with  $z'_0(c_0) = P'_n(c_0) \neq 0$ . Since  $\rho = d^n \prod_j z_j^{d-1}$  and the other cycle points do not vanish at  $c_0$ , the map  $\rho$  vanishes to order exactly  $d-1$  at  $c_0$ . Composing with a Riemann map of  $W$ , we obtain a finite Blaschke product of degree  $d-1$  with a single zero of order  $d-1$ , which is the  $(d-1)$ -st power of a Möbius transformation. For the last statement: along each arc  $|\rho| \rightarrow 1$  and  $\rho \rightarrow -1$ , so the limit set of the arc in  $\partial W$  is connected and consists of parameters at which  $f_c$  has a cycle of period dividing  $n$  with multiplier  $-1$ . These parameters are the zeros of a polynomial in  $c$  that does not vanish at  $c_0$ , so they are finitely many, and the limit set is a point.  $\square$

In degree three we therefore define two bulb sizes  $G_{\pm}$  by Definition 2.4, one at each of the two landing points, and both are predicted. For  $(p, r) = (1, 1)$ ,  $(3, 1)$  and  $(3, 2)$  the predictions agree with the extrapolated bulb values to within  $3 \cdot 10^{-5}$ ; for instance at  $(3, 1)$  the horn map gives 1.196074 and 1.227092, against 1.196075 and 1.227091. For  $(2, 1)$  the two landing points nearly coincide and their labels exchange as  $N$  varies; the sum  $G_+ + G_-$  agrees to  $5 \cdot 10^{-6}$ .

## 9. ARITHMETIC NATURE

For each of  $\kappa_0$ ,  $\kappa_0^{(3)}$  and  $\kappa_{1/2}$  we searched with PSLQ [4], at 28 digits and coefficient bound 1000, for integer relations between the real part, the imaginary part, the squared modulus or the argument divided by  $\pi$ , and every subset of at most three of  $\{1, \pi, \pi^2, 1/\pi, \log 2, \zeta(3), \text{Cat}, \gamma\}$ , where Cat is Catalan's constant. For  $\kappa_0$  we also tried  $\text{Re}(2\pi i \kappa_0)$ ,  $\text{Im}(2\pi i \kappa_0)$ ,  $|a_1|$  and  $\arg a_1/\pi$ . No relation was found. This is weak evidence, since the basis is small and omits, for instance,  $e^\pi$  and  $\Gamma(\frac{1}{4})$ . It is consistent with the expectation that invariants of parabolic germs, such as the Écalle–Voronin moduli [3, 21], have no closed form, and we regard  $\kappa_0$ ,  $\kappa_0^{(3)}$  and  $\kappa_{1/2}$  as new constants of this kind.

## 10. DISCUSSION

The first-order theory of the satellite bulbs is the Ford-circle picture, with radii  $q^{-2}$  and Farey tangencies. The first correction,  $\kappa(p/q)$ , is the holomorphic index of a parabolic germ (Theorem 2.1), and it equals the index of the rotation quotient minus  $\frac{1}{2}$  (Corollary 3.2). Numerically it is controlled by near-parabolic renormalization. The constant  $\kappa_0$  comes from the

simplest instance,  $\alpha = 1/q \rightarrow 0$ . Bounded  $p$  corresponds to one renormalization step with phase  $e^{-2\pi i r/p}$ , and the first order in  $1/q$  comes from the exact Möbius phase. Large partial quotients correspond to iterated renormalization, which contracts by a factor 0.042 per step towards a limit that differs from the one given by the fixed point of Lanford and Yampolsky. Hyperbolicity of  $\mathcal{R}$  would explain the observation, reported in [18], that  $\kappa(p/q)$  depends mainly on the last partial quotients, that is, on  $p^{-1} \bmod q$ . Finally, every rational  $p/q$  carries a pair of such constants: the one-sided limits of  $\kappa$  at  $p/q$  are the indices of the upper and lower parabolic renormalizations of  $e^{2\pi i p/q} z + z^2$ , and for the period-two disc the component enters only through this germ at its root. Farey-type combinatorics survives at every order, while the metric data at a rational are, conjecturally, those of the parabolic germ there.

### Open problems.

- (1) Prove Conjecture 4.5, for instance by carrying out steps (a) and (b) of Section 5.
- (2) Prove Conjecture 5.2, in particular that the renormalized germ has no  $O(1/q)$  correction beyond the phase of Proposition 5.1. By Corollary 5.3 this gives the universal term  $-i/(2\pi pq)$ . Identify the  $O(q^{-2})$  term.
- (3) Prove Conjecture 6.3, prove that the fixed point  $\mathcal{G}_*$  of  $\mathcal{G} \mapsto \mathcal{P}(\bar{\mathcal{G}})$  exists, and describe it and its spectrum.
- (4) Compute the generic limit function  $\kappa(p/q)$ ,  $p \rightarrow \infty$ , by iterated renormalization with bounded partial quotients, including its values at quadratic irrationals as periodic points of  $\mathcal{R}$ .
- (5) Prove Conjecture 7.1; its case  $p/q = 0/1$  is Conjecture 4.5. Relate the jump  $\iota(\mathcal{P}_{p/q}^+) - \iota(\mathcal{P}_{p/q}^-)$  to the arithmetic of  $p/q$ .
- (6) Prove Conjecture 7.3, test it on components of higher period (whose roots have  $q \geq 3$  petals and distinct upper and lower germs), and derive the first-order coefficient  $-0.17743 - 0.19381 i$  of the disc from the kinematics at  $c = -\frac{3}{4}$  (Section 7).
- (7) Decide whether  $\kappa_0$  can be expressed in closed form through the Écalle–Voronin invariants of  $z + z^2$ .

## APPENDIX A. NUMERICAL METHODS AND ERROR CONTROL

**A.1 Normal form.** Recursion (4) was run in `mpmath` [17] at 50 working digits for  $64 \leq q \leq 8192$  ( $p = 1$ ) and for the bounded- $p$  classes of Table 3 ( $q \leq 5124$ ). It was also implemented in Arb ball arithmetic [12]. There the online convolution  $(H^2)_k$  is evaluated in blocks: completed blocks are folded in by truncated series products, and the remainder is summed directly. The working precision is doubled until the certified radius of  $\kappa$  is below  $10^{-40}$ . For  $p = 1$  the radii are  $1.6 \cdot 10^{-50}$  at  $q = 1024$  and below  $2 \cdot 10^{-111}$  at  $q = 32768$ , and the two implementations agree to the 40 digits stored. The ball version is 60–110 times faster;  $q = 32768$  takes about two minutes. For generic  $p$  the coefficients  $H_k$  are very large and ball radii, which cannot see cancellation, grow until they carry no information. For those values we ran the same code in midpoint arithmetic at two precisions, 128 and 192 bits, and took their difference as the error; it never exceeded  $4 \cdot 10^{-35}$ . The recursion for general germs used in Section 6 was run in `mpmath` at 40–60 digits.

**A.2 Extrapolation.** Limits in  $q$  or  $N$  come from least-squares fits by polynomials of degree  $J$  in  $1/q$  or  $1/N$ , solved by QR decomposition at 40–80 digits. We quote the best overdetermined fit and, as its uncertainty, the spread of the fits that remain stable under changes of  $J$  and of the smallest  $q$  used. The limit of the hierarchy (Numerical result 6.1) uses a real-linear model  $\Delta_{k+1} = L\Delta_k$  fitted to the last three differences. This model is exact for both linear and

antilinear geometric contractions, whereas componentwise Aitken extrapolation is not exact for antilinear ones.

**A.3 Horn maps.** For the cusp germs the Fatou series were truncated at  $K = 40$  terms and evaluated at  $|w| \geq 800$ , where the truncation error is below  $10^{-60}$ ; the coefficients  $|c_k|$  grow roughly like  $k!/(2\pi)^k$ . We used  $n = 1000$ – $1800$  repelling iterations and computed Fourier coefficients from 64–80 samples on  $\text{Im } Z = \frac{1}{4}$ , with aliasing below  $10^{-58}$ . For the germs  $\mathcal{G}_k$  of the hierarchy, which are known only as convergent series, the horn maps were sampled at  $\text{Im } Z = 1$  and  $\frac{5}{4}$  so that all orbits stay well inside the disc of convergence. A first run at  $\text{Im } Z = \frac{1}{2}$  was inaccurate at the  $10^{-3}$  level and was discarded. For the germs  $e^{2\pi ip/q}z + z^2$  of Section 7 the Fatou series (exact for  $q = 2$ , Lemma 7.4; at working precision otherwise) was truncated at  $K = 30$ – $60$ , with 600–1500 iterations of  $g$ . The escape band was located by scanning  $\text{Im } Z$  in steps of  $\frac{1}{4}$ , and the samples were taken 1.5 beyond it (at 4.5–5.5 for  $q = 2$ ). The working precision includes the digits lost to the amplification  $e^{2\pi M|h|}$ , and the reproduction of  $\kappa_0$ ,  $\overline{\kappa_0}$  and  $\kappa_{1/2}$  by the general code is part of the test suite. A Fourier coefficient  $a_n$  extracted at height  $h$  carries noise amplified by  $e^{2\pi nh}$ , and the working precision (60–160 digits) was chosen accordingly. Independent heights agree to all digits reported.

**A.4 Multiplier data.** The Taylor coefficients  $r_k$  were computed by Cauchy integrals of  $R_{p/q}$  over a circle, continuing the  $q$ -cycle numerically, and those of the renormalized families in the same way. For  $z^2 + c$  radii up to  $|u| = 2.5$  are admissible. For  $z^3 + c$  the function  $R_{p/q}$  has a singularity inside  $|u| < 2.5$ , and the radius must be at most 1.5. The condition  $r_1 = -1$  of Theorem 2.1 is a reliable diagnostic: on the larger circle it fails, with  $r_1 = -1.032$ . In degree three each satellite has two landing points of multiplier  $-1$  (Proposition 8.1), and both were reached by continuation of the target multiplier from 0 to  $-1$ .

**A.5 Reproducibility.** All code, data and tests are in the accompanying repository [18]. The package `bulbford` contains the normal forms (`normal_form`, `normal_form_ball`), the horn maps (`horn`, `horn_half`), the germ and renormalization tools (`germ`, `renorm`), the extrapolation (`extrapolate`) and the dynamics (`dynamics`, whose cycle iteration is compiled with Numba). Every table and figure is generated from the data files by `paper/tables.py` and `paper/figures.py`. The test suite checks Theorems 2.1 and 3.1, the value of  $\kappa_0$  to 25 digits, the bounded- $p$ , kinematic, hierarchy and degree-three predictions, and the constant  $\kappa_{1/2}$ .

**Data availability.** All data supporting the results are contained in the repository [18].

**Acknowledgments.** The computations and the drafting of this paper were carried out with the assistance of an AI system (Claude, Anthropic).

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